Dr. Fabrizio De Santis

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Parma, Italy



Bibliography

Fabrizio De Santis is a **Ph.D. student in Information Technology at the University of Parma**, where he conducts research at the intersection of process science and artificial intelligence. His work focuses on designing and developing **neuro-symbolic methods** that combine the interpretability of symbolic reasoning with the predictive power of machine learning, with applications spanning **predictive process monitoring** and **clinical decision support** in the healthcare domain. De Santis is driven by the goal of building AI systems that are not only accurate, but also transparent, fair, and reliable in real-world settings. His research has touched on problems such as compliance-aware process monitoring, counterfactual fairness in prescriptive analytics, and sepsis prediction from clinical event data. He earned his Bachelor's degree in Computer, Electronic and Communications Engineering and his Master's degree in Computer Engineering, both from the University of Parma. During his Ph.D., he has collaborated with the **Process and Data Science (PADS) group at RWTH Aachen University**, under the supervision of Prof. Dr. Wil van der Aalst, one of the leading figures in the process mining community.

Professional Experience

University of Parma November 2026 – Present
Research Fellow Parma, Italy

- Neurosymbolic techniques for healthcare and cybersecurity

RWTH Aachen University — PADS Group December 2024 – May 2025
Visiting Ph.D. Student Aachen, Germany

- Research visit under the supervision of Prof. Wil van der Aalst
- Worked on neuro-symbolic AI for predictive process monitoring

University of Parma July 2022 – June 2023
Research Fellow Parma, Italy

- Applied ML and NLP models to the legal domain
- Developed a multi-label classifier for labeling maxims related to the Italian Penal Code
- Fine-tuned a pre-trained BERT model on a large corpus of legal maxims

Teaching

CRPA - Centro Ricerche Produzioni Animali May 2026
Contract Instructor Bologna, Italy

- Topic: Cybersecurity and Generative Artificial Intelligence

ITS Olivetti February 2025 – March 2025
Contract Instructor Parma, Italy

- Topic: Digitalization of processes in business settings and in the creative and cultural sector through AI solutions and immersive technologies

Fondazione Alma Mater November 2024
Contract Instructor Bologna, Italy

- Topic: Services and Innovation for Industry 4.0 - Process Mining and Predictive Process Monitoring

Education

University of Parma November 2023 – October 2026
Ph.D. in Information Technology Parma, Italy

- Topic: Artificial Intelligence for Business Processes
- Thesis Title: Neuro-Symbolic Predictive Process Monitoring: Learning from Data and Domain Knowledge
- Supervisor: Prof. Francesco Zanichelli

University of Parma September 2021 – October 2023
M.Sc. in Computer Engineering' Parma, Italy

- Thesis: Design and development of a recommender system based on NLP and Deep Learning for legal applications
- Supervisor: Prof. Stefano Cagnoni
- Summa Cum Laude

University of Parma September 2017 – April 2021
B.Sc. in Computer, Electronic and Communications Engineering' Parma, Italy

- Thesis: A coevolutionary approach to pattern classification based on Genetic Programming
- Supervisor: Prof. Stefano Cagnoni

Publications

Journal Articles

- F. De Santis, G. Park, F. Zanichelli. **Neuro-symbolic artificial intelligence for real-time sepsis mortality prediction**. *Engineering Applications of Artificial Intelligence*, vol. 177, Part 1, art. 114920, 2026. DOI: 10.1016/j.engappai.2026.114920

Conference Papers

- F. De Santis, G. Park, W.M.P. van der Aalst, F. Zanichelli. **Compliance-Aware Predictive Process Monitoring: A Neuro-Symbolic Approach**. In: *Advanced Information Systems Engineering – 38th International Conference, CAiSE 2026, Verona, Italy, June 8–12, 2026, Proceedings, Part II*. Lecture Notes in Computer Science, vol. 16559, pp. 223–239. Springer, Cham, 2026. DOI: 10.1007/978-3-032-28117-3_13

- F. De Santis, G. Park, F. Zanichelli. **Neuro-Symbolic Learning for Predictive Process Monitoring via Two-Stage Logic Tensor Networks with Rule Pruning**. In: *Advances in Knowledge Discovery and Data Mining – 30th Pacific-Asia Conference, PAKDD 2026, Hong Kong, China, June 9–12, 2026, Proceedings, Part II*. Lecture Notes in Computer Science, vol. 16598, pp. 104–118. Springer, Singapore, 2026. DOI: 10.1007/978-981-92-1462-4_9

- F. De Santis, A. Padella, G. Park, F. Zanichelli, M. de Leoni. **Fairness in Prescriptive Process Analytics: Integrating Fairness Constraints into the Learning Phase**. In: *Business Process Management – 24th International Conference, BPM 2026, Toronto, ON, Canada, September 27–October 2, 2026, Proceedings*. Lecture Notes in Computer Science, vol. 16936, pp. 369–387. Springer, Cham, 2027. DOI: 10.1007/978-3-032-37777-7_23

Workshop Papers

- F. De Santis, F. Zanichelli. **Towards Trustworthy Predictive Process Monitoring: A Neuro-Symbolic Perspective**. Presented at *AI4PMMA 2026 – International Workshop on AI for Process Management, Mining, and Analytics*, co-located with the 24th International Conference of the Italian Association for Artificial Intelligence (AIxIA 2026), Perugia, Italy, October 6–9, 2026. To appear in CEUR Workshop Proceedings.

- A. Padella, F. De Santis, F. Zanichelli, M. de Leoni. **Performance and Human Perspective in Prescriptive Process Analytics: Advances and Open Challenges**. Presented at *AI4PMMA 2026 – International Workshop on AI for Process Management, Mining, and Analytics*, co-located with the 24th International Conference of the Italian Association for Artificial Intelligence (AIxIA 2026), Perugia, Italy, October 6–9, 2026. To appear in CEUR Workshop Proceedings.

Tutorials

- G. Park, F. De Santis. **Neuro-symbolic AI for BPM: A Tutorial on Predictive and Prescriptive Process Analytics**. In: *Business Process Management – 24th International Conference, BPM 2026, Toronto, ON, Canada, September 27–October 2, 2026, Proceedings*. Lecture Notes in Computer Science, vol. 16936. Springer, Cham, 2027. DOI: 10.1007/978-3-032-37777-7_4

Skills

Languages : Python, JavaScript, HTML/CSS, SQL, C, C++, Java

Libraries : PyTorch, scikit-learn, Pandas, Numpy, Hugging Face Transformers/TRL, vLLM, Unsloth, LTNtorch

Topics : Machine Learning, Neuro-Symbolic AI, Predictive Process Monitoring, Healthcare AI, LLM Fine-Tuning (LoRA, DPO, GRPO), Natural Language Processing

Languages

Italian Mother tongue

English B2